

Implementation Plan

Phase 1: Project Analysis

Deliverables

- Understand current architecture
 - Review existing nodes
 - Identify duplicated code
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Phase 2: Node Abstraction

Build:

BaseNode Component

Refactor:

- Input Node
- Output Node
- LLM Node
- Text Node

Deliverable:

Single reusable architecture

Phase 3: Additional Nodes

Create:

1. API Node
2. Database Node
3. Condition Node
4. Notification Node
5. Math Node

Deliverable:

Five working nodes using BaseNode

Phase 4: Styling

Implement:

- Design system
- Color palette
- Typography
- Node styling
- Button styling

Deliverable:

Production-quality UI

Phase 5: Text Node Enhancements

Implement:

- Auto-sizing width
- Auto-sizing height
- Variable detection
- Dynamic handles

Testing:

- Multiple variables
 - Variable removal
 - Duplicate variables
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Phase 6: Backend Integration

Frontend:

- Update submit.js

Backend:

- Update parse endpoint

Deliverable:

Connected system

Phase 7: DAG Validation

Implement:

- Graph construction
- DFS cycle detection

Deliverable:

Accurate DAG response

Phase 8: User Feedback

Implement:

- Success modal
- Error modal
- Loading state

Deliverable:

Professional user experience

Phase 9: Testing

Test:

- Node creation
 - Node connections
 - Variable parsing
 - API integration
 - DAG validation
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Phase 10: Final Submission

Review:

- Code quality
- UI consistency
- Performance

Prepare:

- GitHub repository
- Demo recording
- Submission package